CTS
Centre for Technology and Society
UNIVERSITY OF OXFORD

WORKING PAPER SERIES
WP 2026-05

The Crisis of the Tax State Revisited

Artificial Intelligence, Business Conflict, and Digital Fiscal Capacity

Jacob Edenhofer
Nuffield College, University of Oxford

Ben Ansell
Nuffield College, University of Oxford

NOVEMBER 2026

ABSTRACT

Rich democracies finance the modern state through labour taxation. If artificial intelligence (AI) reduces the labour share, governments will have to replace lost revenue, especially through consumption or capital taxation. AI rents arise from intangible assets with hard-to-observe transaction prices. Taxing them requires digital fiscal capacity, the legal and administrative resources for identifying and valuing AI returns and distinguishing rents from normal returns. In its absence, blunt capital taxation disproportionately burdens conventional, non-AI firms. With digital fiscal capacity, targeted taxation becomes possible and shifts part of the burden to AI firms, making conventional firms pivotal in the conflict between voters favouring such taxation and AI firms opposing it. Our model examines investment in this capacity as the transition to an AI economy empowers AI firms politically. Declining revenue from labour taxation and greater inequality strengthen conventional firms' support; their fiscal stakes, the speed of transition, and political power relative to AI firms determine whether and when investment occurs. The extent to which technological change transforms the tax state thus depends crucially on intra-business conflict over the incidence of capital taxation.

artificial intelligence · taxation · fiscal capacity · business conflict

Working papers are circulated for discussion and comment; they have not been peer reviewed. Views expressed are those of the authors.

The Crisis of the Tax State Revisited

Artificial Intelligence, Business Conflict, and Digital Fiscal Capacity

Jacob Edenhofer
University of Oxford*

Ben Ansell
University of Oxford

September, 2026

Abstract

Rich democracies finance the modern state through labour taxation. If artificial intelligence (AI) reduces the labour share, governments will have to replace lost revenue, especially through consumption or capital taxation. AI rents arise from intangible assets with hard-to-observe transaction prices. Taxing them requires digital fiscal capacity, the legal and administrative resources for identifying and valuing AI returns and distinguishing rents from normal returns. In its absence, blunt capital taxation disproportionately burdens conventional, non-AI firms. With digital fiscal capacity, targeted taxation becomes possible and shifts part of the burden to AI firms, making conventional firms pivotal in the conflict between voters favouring such taxation and AI firms opposing it. Our model examines investment in this capacity as the transition to an AI economy empowers AI firms politically. Declining revenue from labour taxation and greater inequality strengthen conventional firms' support; their fiscal stakes, the speed of transition, and political power relative to AI firms determine whether and when investment occurs. The extent to which technological change transforms the tax state thus depends crucially on intra-business conflict over the incidence of capital taxation.

*Edenhofer: University of Oxford, Nuffield College. Email: jacob.edenhofer@nuffield.ox.ac.uk. Ansell. University of Oxford, Nuffield College. Email: ben.ansell@politics.ox.ac.uk.

1 Introduction

Schumpeter (1991, 101) famously argued that changes in economic organisation can produce "a crisis of the old fiscal methods" and ultimately the tax state itself. In this paper, we examine the political implications of the rise of (generative and agentic) artificial intelligence (AI) in light of Schumpeter's observation. Rich democracies rely heavily on taxing labour income to finance the modern state, principally through personal income taxes and social security contributions (Keen and Slemrod, 2022). For all the uncertainties[1] surrounding the rise of AI—relating to debates about its consequences for growth, employment, inequality, market power, and polarisation (Furman and Seamans, 2019; Jones and Tonetti, 2020; Boix, 2024; Kurz, 2023; Kasy, 2025, 2026; Restrepo, 2025; Jones, 2026; Gratton and Lee, 2026; Acemoglu et al., 2026b)—it seems certain it will put pressure on the political stability of this arrangement. As AI diffuses across the economy, the share of labour income will decline (Korinek and Lockwood, 2026), forcing governments to raise revenue via other taxes.

From the perspective of public finance scholarship, it is relatively straightforward to "optimally" adjust the fiscal structure of the state in response to that challenge. Korinek and Lockwood (2026) argue that consumption taxation should become the principal revenue instrument as labour's share declines, supplemented by taxes on "AI rents"—those profits that exceed the risk-adjusted return.[2] However, this addition of taxing "AI rents" is not innocuous. Most governments in rich democracies currently have the capacity to assess taxable labour income and reported corporate profits, but they struggle to identify AI. This is because the data, software, algorithms, and organisational capital generating these rents are often internally produced, lack observable transaction prices, and can generate profits across legal entities and jurisdictions (Veldkamp, 2023; Farboodi and Veldkamp, 2023, 2026). They are what Haskel and Westlake (2018) call *intangible assets*. Taxing them requires governments to build what we will call *digital fiscal capacity* (Healy et al., 2026; Fetzer et al., 2024), understood here as the legal and administrative systems needed to identify and value AI-related returns and to distinguish rents from

[1] E.g, the disagreement over aggregate productivity (Acemoglu, 2025; Aghion and Bunel, 2024).

[2] A fixed consumption tax is equivalent to a combination of taxes on labour income and initial wealth. As it applies at the same rate in every period, it leaves the relative price of present and future consumption unchanged and does not distort saving. Taxing pure rents likewise does not discourage investment because investors retain the risk-adjusted return required to induce the activity.

normal returns.

While the importance of building up this capacity is increasingly recognised (e.g. Korinek and Lockwood, 2026), it is too often seen as a merely technical decision. Taking our cue from work in historical political economy on what, since Scott (1999), has become known as *legibility*—the state's ability to render people, assets, and economic activity observable and classifiable for administrative purposes—we argue that the decision whether to invest in digital fiscal capacity is instead deeply political. Rendering new assets and income visible to the state redistributes tax burdens from conventional firms, whose tangible assets and reported profits are already readily assessed, towards the ascendant firms earning AI rents, giving rise to distributional conflict (Scott, 1999; Lee and Zhang, 2017; Brambor et al., 2020; Garfias and Sellars, 2026). Whoever prevails in this conflict determines how Schumpeter's "crisis of the old fiscal methods" is resolved. Fiscal innovation allows the state to tax the ascendant firms of the AI economy; failure to innovate leaves old methods in place, concentrating the burden on consumers and on those firms whose resources remain visible, even as their political influence declines. In other words, whether digital fiscal capacity is created is, as with many historical cases (Ansell and Samuels, 2010), a conflict among elites.

Our argument—which we formalise in an infinite horizon Markov model—identifies conventional business, the non-AI sector, as pivotal to how this conflict is resolved. We start from the assumption that public expenditure is constant, implying that any erosion of revenue from labour taxation creates a residual fiscal gap. We begin with a simplifying electoral assumption: voters determine how much of this gap is financed by taxes on consumption and how much by taxes on capital.[3] Greater inequality increases the contribution that they prefer capital to make. Firms earning AI rents prefer the burden to fall on consumption and therefore resist investments that would make their returns easier to tax.

Electoral demand for a capital contribution does not, however, determine which firms pay it. When *digital fiscal capacity* is low, the state cannot identify AI rents, instead funding the capital contribution through blunt capital taxation. Because conventional firms have relatively tangible assets and readily assessed profits, they

[3]This starting point connects to comparative political economy research on the political foundations of the tax structure. Cusack and Beramendi (2006) discuss how wage coordination, partisanship, and political institutions contributed to reliance on labour taxation, while Beramendi and Rueda (2007) show how corporatist commitments can induce social democratic governments to finance the welfare state through regressive consumption taxation (see also Andersson, 2022).

bear a disproportionate share of this burden. Building, digital fiscal capacity allows the state to replace part of the blunt levy with targeted taxation of AI rents. Conventional firms therefore support investment in digital fiscal capacity when the resulting reduction in their expected tax burden exceeds the political cost of inducing the government to invest (through contributions, lobbying, etc.). Their prospective burden under blunt taxation, and hence incentive to demand digital fiscal capacity, increases both when the erosion of revenue from labour taxation raises the residual fiscal gap and when inequality raises the share of tax revenue that voters want capital to fund.

The argument also has a dynamic component. Building investment in digital fiscal capacity imposes an immediate political cost on conventional firms, while the resulting capacity becomes effective only after a delay and then persists. What determines timing is how soon there will be a transition to a persistent AI-intensive environment—we might think of this as a transition to Artificial General Intelligence that replaces large shares of current labour. In this AI-intensive world, lower revenue from labour taxation creates a large residual fiscal gap, inequality is greater, and political influence shifts towards AI firms.

We conceive of two types of investment in digital fiscal capacity driven by conventional firms in response. Investment before the transition to an AI-intensive environment is *anticipatory*; investment after it is *reactive*. Anticipatory investment secures capacity while conventional firms retain political influence, but requires them to bear the political cost before receiving any tax benefit. Waiting postpones this cost, but leaves them exposed to blunt taxation after the transition and makes investment harder as their political influence declines. A faster expected transition or a larger prospective burden under blunt taxation therefore favours anticipatory investment. If conventional firms retain sufficient influence after the transition, intermediate fiscal stakes produce reactive investment. However, this depends on the political balance of power in an AI-powered world. A large loss of relative political influence for conventional firms eliminates the possibility of reactive investment in digital fiscal capacity, leaving anticipatory investment at high stakes and no investment at low stakes.

These results identify the political conditions under which the public-finance prescription for the AI transition can be realised. The intuitive result is that the erosion of labour-tax revenue and greater inequality increase demand for digital fiscal capacity through distinct channels. The former enlarges the residual fiscal

gap, while the latter raises the share of that gap voters want capital to finance. Both increase conventional firms' expected burden under blunt capital taxation and therefore their willingness to support investment in digital fiscal capacity.

The less intuitive result concerns conventional firms' prospective loss of political influence. The growing power of AI firms does not uniformly obstruct fiscal innovation. When fiscal stakes are high, the prospect of losing political influence induces conventional firms to invest before the transition. A faster expected transition strengthens this incentive by bringing the tax benefits closer. At intermediate stakes, firms that retain sufficient influence instead invest reactively. A larger loss of political influence removes this possibility, leaving anticipatory investment at high stakes but persistent low capacity at lower stakes. An anticipated loss of political influence can therefore create a constituency for updating Schumpeter's old fiscal methods or eliminate it before it acts. Accordingly, the model here mirrors the logic of preventive war in the international relations literature—facing a new rival, an actor must decide whether to bear onerous costs now, given the possibility of not having sufficient relative power later (Fearon, 1995).

While our model draws on existing analyses of dynamic competition, it has a number of aspects that speak directly to the challenge of an AI transition. First, AI erodes labour-tax revenue, forcing new sources of revenue. Second, AI increases inequality, shifting popular opinion towards higher taxation of capital. Third, AI produces rents among AI-firms that depend on difficult-to-value and hence difficult-to-tax intangible assets. Fourth, AI likely shifts political influence towards AI firms earning those rents and away from conventional firms. AI therefore combines greater pressure for capital to contribute with rents that existing fiscal systems struggle to assess along with new firms whose growing political influence allows them to resist investment in that fiscal capacity. The speed of these concurrent changes determines whether conventional firms can establish digital fiscal capacity before their relative influence declines.

Typically, research on state capacity typically focuses on investment in classic fiscal and legal institutions (Besley and Persson, 2009, 2011; Mares and Queralt, 2020; Li et al., 2022), with limited attention to their digital counterparts (for exceptions, see Fetzer et al., 2024; Healy et al., 2026). A related literature locates changes in state capacity, broadly construed, in conflict between established and ascendant economic groups (Ansell and Samuels, 2010, 2015; Beramendi et al., 2016; Garfias, 2018). Our account connects these arguments through the incidence of

capital taxation. The lack of digital fiscal capacity imposes a disproportionate burden on conventional firms and thereby creates business support for investment in the state's ability to tax an ascendant sector. This extends accounts of intra-elite conflict to the adjustment of fiscal institutions as economic organisation changes. It also yields the dynamic implication that an anticipated decline in the influence of conventional business can accelerate investment in capacity when firms expect continued administrative weakness to expose them to sufficiently high taxes.[4]

The emerging political economy literature on AI has concentrated on political outcomes differing from those studied here. One strand examines surveillance, repression, and relations between states and firms in autocracies (Beraja et al., 2023a,b; Acemoglu et al., 2026a; Yang, 2026). Another strand studies how occupational exposure to AI and subjective perceptions of employment risk (see also Thewissen and Rueda, 2019) influence support for redistribution, compensation, regulation, or AI itself (Grant et al., 2025; Haslberger et al., 2025; Lall, 2025; Magistro et al., 2026; Mansell et al., 2026). We set aside the former as our analysis concerns rich democracies. The latter shows how citizens form political demands in response to AI, which our model captures in reduced form, through greater demand for taxation of capital relative to consumption, in a manner similar to Meltzer and Richard (1981). Unlike in Meltzer and Richard, however, in our model public expenditure remains fixed, meaning the relevant choice concerns the composition of taxation—consumption vs capital taxation, and broad vs targeted capital taxation—rather than the size of government. Core for us, rather than voters *per se*, is the relative political influence of conventional firms in the shadow of an AI transition.

The next section presents the model, characterises the equilibrium regimes, and derives the comparative statics that are central to the argument. The subsequent discussion considers the interpretation, scope, and limitations of the mechanism. The final section concludes.

[4] Callander (2026) conceptualises AI regulation as a race between policy learning and the emergence of an interest group representing the AI industry. Industry development improves policymakers' information while allowing AI firms to convert market power into political influence, which creates an incentive for early delegation. Our account abstracts from learning, instead locating the incentive for early action in tax incidence. Conventional firms support capacity because blunt capital taxation burdens them disproportionately, while an expected loss of relative power raises the future cost of securing it. This resembles the preventive logic in Fearon (1995), in which an expected deterioration in relative power can induce costly action before it occurs. Fiscal stakes determine whether conventional firms act on this incentive.

2 Formalising the argument

The model in this section formalises when conventional firms support investment in the capacity required to tax AI rents. We begin by setting out the idea of two states of the world—before and after an AI transition. We then derive the fiscal gap that needs filling once labour taxation declines. We then shift to setting out the level of capital taxation versus consumption taxation preferred by voters. We then move to the core of the model—the decision to invest in digital fiscal capacity and how that alters the relative tax burden of conventional firms versus AI firms. The remainder of the model introduces the relative political influence of conventional firms and derives their dynamic equilibrium choice over whether to engage in costly lobbying for the state to invest in digital fiscal capacity, along with whether the timing of that investment is anticipatory or reactive.

2.1 Formal setup

The model zeroes in on the conflict over the state's ability to distinguish difficult-to-assess AI rents from the capital income of firms that are already easy to assess. An AI transition creates a fiscal shortfall and may increase inequality. Electoral pressure means that part of that shortfall has to be funded by taxing capital. In the absence of the requisite fiscal capacity, the state collects a disproportionate share of the capital contribution from conventional firms via blunt capital taxation. Those firms may consequently support investments that allow the state to identify and tax AI rents more accurately. The AI transition may also shift relative political power towards AI firms, which raises the cost of investing in capacity after it occurs.

The logic of the model is that the properties of the AI transition determine the fiscal shortfall and its politically required division between consumption and capital taxation. Existing fiscal capacity determines the division of the capital contribution between conventional firms and AI firms.

The AI transition and the fiscal requirement In our model, time is discrete and infinite, with periods indexed by $t \in \mathbb{N}_0$. There are two economic states s_t of the world, depending on whether we are before or after the AI transition:

$$s_t \in \mathcal{S} \equiv \{\ell, h\},$$

where ℓ denotes the economy before the AI transition and h denotes the economy after the transition. The initial state is $s_0 = \ell$. The state evolves according to

$$P = \begin{pmatrix} 1-\pi & \pi \\ 0 & 1 \end{pmatrix}, \qquad \pi \in (0,1], \tag{1}$$

where the rows and columns are ordered as (ℓ, h). Thus

$$\mathbb{P}(s_{t+1} = h \mid s_t = \ell) = \pi, \qquad \mathbb{P}(s_{t+1} = h \mid s_t = h) = 1.$$

Let

$$\tau \equiv \inf\{t \geq 1 \mid s_t = h\}$$

denote the date of the transition. The random variable τ is geometrically distributed[5] and

$$\mathbb{E}[\tau \mid s_0 = \ell] = \frac{1}{\pi}. \tag{2}$$

The parameter π therefore measures the speed of the transition. Greater values mean that the expected period during which capacity can be acquired in the low state is shorter.

The two-state process amounts to conceiving of the AI transition as a change between two fiscal environments. This is mainly a simplifying assumption. Assuming the high state to be absorbing is a more substantive analytical commitment; it means the fiscal and political consequences of the transition are persistent ad infinitum once the high state is reached. Some of these assumptions might be innocuous - if AGI does happen, it seems likely the labour share collapse will be permanent. Some are less innocuous, since we assume once digital fiscal capacity is established it is permanent. This rules out technological reversals, or efforts by AI companies to roll back or undermine investments in state capacity.

We now move to discussing the fiscal gap created by a collapse in the labour share due to widespread AI. We begin by assuming that there is a fixed level of

[5]Its support is $\{1, 2, \ldots\}$, with $\mathbb{P}(\tau = n) = \pi(1-\pi)^{n-1}$ and $\mathbb{P}(\tau > n) = (1-\pi)^n$. The geometric distribution is "memoryless" since $\mathbb{P}(\tau > t+k \mid \tau > t) = (1-\pi)^k = \mathbb{P}(\tau > k)$. Conditional on the economy remaining in state ℓ, the probability of transition in the next period is therefore always π, regardless of the time already spent in that state. This property makes the investment problem in state ℓ stationary and means that elapsed time provides no additional information about the transition date. It excludes learning and duration dependence in the transition process, either of which would require beliefs or elapsed time to enter the state space.

public expenditure that must be met by tax revenues, accordingly we abstract away from changes in overall spending demands. Let $\bar{G} > 0$ denote committed public expenditure, and let R_s^0 denote revenue from the existing tax system, including labour taxation. The residual fiscal requirement is

$$S_s \equiv \bar{G} - R_s^0. \tag{3}$$

We impose the following assumptions:

$$R_\ell^0 = \bar{G}, \qquad R_h^0 = \bar{G} - S, \qquad S_\ell = 0, \qquad S_h = S > 0. \tag{4}$$

The parameter S, which reflects the fiscal requirement following the transition, thus captures the eventual fiscal severity of the transition, while π measures its speed. The assumption $S_\ell = 0$, means that in the pre-AI transition world, budgets are balanced, and this rules out any current fiscal benefit from capacity before the transition. Investment in state ℓ must therefore be justified entirely by expected future savings. A positive low-state requirement would make early investment more attractive without changing the source of the incentive.

The fiscal requirement is financed through the two types of taxes, taxes on consumption or capital, as we will discuss below. We hold public expenditure constant and thus exclude borrowing or (sustained) reductions in expenditure (e.g. large-scale austerity).[6] This assumption ensures that the erosion of the existing tax base creates a residual fiscal requirement, i.e. to make up for the shortfall in labour tax revenue that is required for funding the constant level of expenditure by increasing the tax burden on either consumption or capital.

Let i_s denote inequality in state s, and assume

$$i_h \geq i_\ell. \tag{5}$$

The weak inequality allows for AI transitions that erode the labour tax base without increasing inequality. The parameters S and i_h vary independently and represent the fiscal and distributional severity of the high state. Linking both to the state of

[6] A more general expression is: $S_s = \bar{G} - R_s^0 - D_s - E_s - L_s$, where D_s is borrowing, E_s is an expenditure reduction, and L_s is additional labour-tax revenue. The baseline determines these terms at zero. Full adjustment through any one of them would remove the demand for the consumption and capital contributions studied here. The model also excludes debt dynamics and intertemporal tax smoothing.

the world $s_t \in \{s_l, s_h\}$) avoids a separate inequality process. It does require their timing to coincide. An earlier or otherwise temporally asynchronous increase in inequality would create electoral pressure for a change in the tax mix before the fiscal shortfall appears. This could be explored in an extension of the model.

The politically required tax mix Given a fiscal requirement of S_s, how can this be financed? The government finances S_s through a consumption contribution C_s and a capital contribution K_s. How are these contributions determined? We argue that there is a reduced-form policy preference, determined by voters, that produces the preferred tax mix. A share α of S_s comes from capital K_s , with $1 - \alpha$ coming from consumption C_s. The chosen level of α is a function of inequality i. Let

$$\alpha \colon \mathcal{D} \to [0, 1]$$

be continuously differentiable on the admissible set of inequality levels $\mathcal{D} \subseteq \mathbb{R}$, with the preferred share α of the fiscal requirement coming from capital taxation rising in inequality:

$$\alpha'(i) \geq 0. \tag{6}$$

The politically required tax mix is

$$K_s = \alpha(i_s) S_s, \tag{7}$$

$$C_s = \big[1 - \alpha(i_s)\big] S_s. \tag{8}$$

The two contributions satisfy $K_s + C_s = S_s$. Working with revenue contributions avoids the need to specify tax bases and behavioural elasticities in the baseline. Recall that budgets are balanced in the pre-transition period $S_l = 0$, therefore $K_l = C_l = 0$ and hence there is no need for capital or consumption taxation. In the post-transition period, $S_h = S$ and so we have $K_h = \alpha(i_h)S$ and $C_h = [1 - \alpha(i_h)]S$.

Equation (6) specifies the electoral link between inequality and tax composition in a reduced form. The link follows the standard Meltzer and Richard (1981) logic. Lower-income households commonly consume a larger fraction of current income and own a smaller fraction of taxable capital than wealthier households. If inequality increases the difference between the pivotal voter's exposure to consumption taxation and their exposure to capital taxation, then their politically preferred capi-

tal share of taxation α rises. While this assumption may fail,[7] we think it is plausible in most rich democracies, where capital ownership is concentrated among the rich, meaning that a capital tax is mainly borne by the latter while a broad consumption tax imposes a larger burden relative to income on the pivotal voter who does not own any significant amount of capital.

The function $\alpha(i)$ summarises electoral competition in reduced form, though it can be thought of as, for instance, representing the preference of a pivotal voter or the outcome of probabilistic voting setup. Whatever the exact microfoundation, voters determine the aggregate contribution required from capital. They do not determine its allocation across firms. The model therefore retains the Meltzer–Richard intuition that greater inequality increases demand for redistribution while concentrating on the composition of revenue used to meet a given fiscal requirement.

Fiscal capacity and the incidence of capital taxation We now turn to the core of our model: the preferences of firms over how capital should be taxed. We model an economy with two sectors. Sector N contains conventional firms with relatively tangible assets, established domestic operations, and tax bases that the state can already assess. Sector A contains AI firms whose returns rely more heavily on intangibles, data, algorithms, organisational capital, and rents whose legal or economic location is difficult to establish. The central difference between both sectors is the ease with which their rents can be taxed, which we—following the literature on intangibles (Haskel and Westlake, 2018; Brynjolfsson et al., 2021; Veldkamp and Baley, 2025; Veldkamp, 2023)—assume to be easier for the conventional than the AI sector.

Let

$$a_t \in \{0, 1\}$$

denote the capacity available at the beginning of period t. The value $a_t = 0$ denotes blunt capital taxation, under which the state cannot attribute AI rents accurately. The value $a_t = 1$ denotes targeted capacity, under which the state can identify the relevant rents, determine liability, and enforce the associated tax treatment.

Let $\zeta \in (0, 1)$ be the share of the capital contribution borne by conventional firms

[7] Consumption-tax exemptions, broad capital ownership across the income distribution, policies facilitating the latter (Atkinson, 2015; Ranaldi, 2026), or the pass-through of capital taxes to wages can weaken or reverse the assumed relationship. Incorporating this into the analysis would make for an interesting extension.

when capacity is absent. Let $r \in (0, \zeta)$ be their share when capacity is available. Define

$$\chi \equiv \zeta - r > 0. \tag{9}$$

The sectoral burdens are then given by

$$T_N(s, a) = \big[\zeta - a\chi\big] K_s, \tag{10}$$

$$T_A(s, a) = \big[1 - \zeta + a\chi\big] K_s. \tag{11}$$

Hence

$$T_N(s, a) + T_A(s, a) = K_s. \tag{12}$$

It is worth briefly outlining what these burdens look like depending on whether the AI transition has occurred (s_l or s_h) and whether investment in digital fiscal capacity has been made ($a = 0$ or $a = 1$). In the pre-transition world, budgets are balanced, $S_l = 0$ and hence $K_l = 0$ and $T_N(s_l, 0) = T_A(s_l, 0) = 0$. Tax burdens only differ once the transition has occurred. If there has not been investment in digital fiscal capacity, capital taxation is blunt and the tax burden for the convention sector is $T_N(s_h, 0) = \zeta K_h = \zeta\alpha(i_h)S$. The tax burden for the AI sector would be $T_A(s_h, 0) = (1 - \zeta)K_h = (1 - \zeta)\alpha(i_h)S$. If there has been investment in digital fiscal capacity then the relative tax burdens of each sector following the AI transition change to $T_N(s_h, 1) = (\zeta - \chi)K_h = (\zeta - \chi)\alpha(i_h)S$ and $T_A(s_h, 1) = (1 - \zeta + \chi)K_h = (1 - \zeta + \chi)\alpha(i_h)S$.

Accordingly, greater fiscal or digital state capacity effectively reallocates the tax burden χK_s from conventional firms to AI firms while the aggregate capital contribution remains constant. The incidence wedge χ captures the lower informational demands of taxing conventional firms. Blunt taxation concentrates the burden on observable assets, payroll, accounts, and legal entities. Capacity for targeted taxation, by contrast, enables the state to attribute a larger share of the contribution to AI rents.[8]

[8] Let $B_j > 0$ denote the readily observable tax base of sector $j \in \{N, A\}$, and let $R_j > 0$ denote its tax-relevant rent. If blunt taxation allocates the capital contribution in proportion to observable bases and targeted taxation allocates it in proportion to rents, then

$$\zeta = \frac{B_N}{B_N + B_A}, \qquad r = \frac{R_N}{R_N + R_A}.$$

The inequality $r < \zeta$ is equivalent to $R_A/B_A > R_N/B_N$. AI firms then have more tax-relevant rent per unit of readily observable base.

Equation (12) separates the electoral choice of the capital contribution from its allocation. Low digital fiscal capacity, which the policymaker has to make do with in the absence of investments, does not eliminate capital taxation. It makes conventional firms the residual source of part of the required revenue. Substantively, this assumption entails that governments *cannot* replace all of the uncollected AI tax with additional consumption taxation, borrowing, or expenditure reductions. The main results still hold if we relax this assumption—as long as a positive residual burden remains with conventional firms.

The current saving that capacity provides to conventional firms is

$$x_s \equiv T_N(s,0) - T_N(s,1) = \chi K_s = \chi \alpha(i_s) S_s. \tag{13}$$

Equations (4) and (13) imply (presuming there is at least some capital taxation, or $\alpha(i_h) > 0$).

$$x_\ell = 0, \qquad x_h = X, \qquad X \equiv \chi \alpha(i_h) S > 0, \tag{14}$$

The scalar X will be our core focus in terms of the incentives of conventional firms. It is the high-state (i.e. post transition) benefit that conventional firms obtain from digital fiscal capacity. X combines the fiscal shortfall S, the politically required capital share $\alpha(i_h)$, and the incidence wedge χ. In sum, we focus on the distributional consequences of investments in digital fiscal capacity between the conventional and AI sector ($x_h > 0$), with greater capacity reallocating the tax burden from the former to the latter.

Investment in digital fiscal capacity and the role of relative political power We now turn to discussing the development of digital fiscal capacity and its relationship to the relative political power of conventional firms.

The initial digital fiscal capacity stock is $a_0 = 0$. When capacity is absent, the conventional sector may induce the government to invest in it. Let $I_t \in \{0,1\}$ denote the investment decision, with the feasible action set

$$\mathcal{I}(a) = \begin{cases} \{0,1\}, & a = 0, \\ \{0\}, & a = 1. \end{cases} \tag{15}$$

Capacity evolves according to

$$a_{t+1} = a_t + (1 - a_t) I_t. \tag{16}$$

An investment made in period t first affects taxation in the following period $t + 1$, hence it is not immediate. Once established, capacity remains available and is not reversible.

The binary choice is mainly a simplifying assumption that keeps the analysis simple.[9] Substantively, it captures a reform with fixed cost or a minimum effective scale, such as a specialist tax unit, a reporting system, or a statutory category for AI rents. A continuous capacity stock would introduce considerations relating to imperfect assessability, which are interesting but also detract from the central tension that the model is designed to elucidate. Similarly, the one-period lag lag reflects the time required to legislate rules, collect information, recruit specialists, and establish enforcement procedures.

The assumption about durability, by contrast, has more substantive importance. On the one hand, it implies that institutions, legal authority, and administrative routines persist after their creation, which is consistent with the arguments relating to path dependence. On the other hand, assuming durability also makes explicit that our model abstracts from commitment problems. Once capacity has been established, the high-state political environment cannot abolish it or prevent its use. Relative political power affects investment in absent capacity, while existing capacity remains effective. If AI firms can dismantle the institution or induce the government to leave it unused, early investment provides weaker protection.

The conventional sector must incur a political cost to induce investment in fiscal capacity. Let $c > 0$ denote this cost before the transition. Let $\rho \in (0, 1]$ index the sector's political power relative to AI firms after the transition, with its pre-transition relative power normalised to one. We assume that the effective cost of inducing investment is inversely proportional to relative political power. Hence

$$k_\ell = c, \qquad k_h = \frac{c}{\rho}. \tag{17}$$

[9] A continuous alternative can use $a_t \in [0, 1]$, the law $a_{t+1} = (1 - \delta) a_t + I_t$, and a convex investment cost. Depreciation would reduce the value of early investment. The baseline sets depreciation to zero and gives capacity a one-period development lag. These restrictions make the parameter regions transparent and give early investment a clear institutional interpretation.

Accordingly for any $\rho < 1$, the political costs of inducing investment in digital fiscal capacity are higher in the post-transition period than they are in the pre-transition period ($k_h > k_l$). A lower ρ represents a larger shift in relative political power towards AI firms and raises the cost of inducing investment after the transition. Appendix A provides a mobilisation technology that gives rise to equation (17).

We assume that the economic transition and the shift in relative political power occur simultaneously. This restriction keeps the state space small and expresses the claim that the growth of AI-linked rents and firms can increase their ability to resist taxation. It excludes independent political shocks and gradual changes in relative political power within each economic state.

The conventional sector acts as a unitary political actor and internalises the aggregate reduction in its tax burden. This assumption removes coordination and free-riding within the sector. Fragmentation can be represented by a larger c or by an additional coordination game. The government is also absent as an independent strategic actor. The reduced form treats investment as occurring when conventional firms incur the relevant political cost. Independent governmental preferences could create an additional motive for investment or allow the government to refuse the subsequent use of capacity.

Timing and payoffs Each period has four stages. At the beginning of period t, the actors observe (s_t, a_t). The government raises S_{s_t}, and the current capital contribution is allocated according to a_t. When $a_t = 0$, the conventional sector chooses mobilisation effort and pays its cost. Successful mobilisation determines $I_t = 1$. At the end of the period, the economic state changes according to P, and capacity changes according to equation (16). Capacity established through investment in period t is available when taxes are collected in period $t + 1$.

The conventional sector's current payoff, apart from profits that are constant across capacity choices, is

$$-T_N(s_t, a_t) - k_{s_t} I_t = -\zeta K_{s_t} + a_t x_{s_t} - k_{s_t} I_t. \tag{18}$$

The term $-\zeta K_{s_t}$ is independent of the investment decision, and that decision does not affect the transition of s_t. Normalising this term to zero gives

$$u(s_t, a_t, I_t) = a_t x_{s_t} - k_{s_t} I_t. \tag{19}$$

The normalisation changes payoff levels and leaves every investment comparison unchanged.[10]

The conventional sector discounts future payoffs by $\beta \in (0,1)$ and chooses a feasible sequence of investment decisions to maximise

$$U_N = \mathbb{E}_0 \left[\sum_{t=0}^{\infty} \beta^t \left(a_t x_{s_t} - k_{s_t} I_t \right) \right]. \tag{20}$$

The payoff-relevant state is

$$z_t \equiv (s_t, a_t) \in \mathcal{Z} \equiv \mathcal{S} \times \{0,1\}. \tag{21}$$

There are four formal states,

$$(\ell,0), \quad (\ell,1), \quad (h,0), \quad (h,1).$$

Only $(\ell,0)$ and $(h,0)$ admit an investment decision. Conditional on $I \in \mathcal{I}(a)$, the joint transition kernel is

$$\mathbb{P}\big(z_{t+1} = (s',a') \mid z_t = (s,a), I_t = I\big) = P_{ss'} \mathbf{1}\big\{a' = a + (1-a)I\big\}. \tag{22}$$

Equations (20) and (22), together with the feasible action set in equation (15), define the dynamic political problem.

2.2 Equilibrium

We now turn to characterising the equilibrium. Appendix B provides the proofs and additional derivations.

Equilibrium concept A stationary Markov strategy is a family

$$\sigma = \{\sigma(\cdot \mid s,a)\}_{(s,a)\in\mathcal{Z}}, \qquad \sigma(\cdot \mid s,a) \in \Delta\big(\mathcal{I}(a)\big) \quad \text{for every } (s,a) \in \mathcal{Z}. \tag{23}$$

[10] The normalisation would cease to be innocuous if investment in capacity affected AI adoption, the size of the tax base, or the transition probability π. In those cases, the omitted term could vary across policies through the endogenous economic state.

The strategy depends only on the current economic state and the current capacity stock. Since the conventional sector is the only strategic actor in the reduced model, sequential optimality at every state provides the relevant notion of perfection.

Definition 1. A *stationary Markov perfect equilibrium* is a stationary Markov strategy σ^* that maximises the conventional sector's continuation payoff at every $(s, a) \in \mathcal{Z}$, given the transition kernel in equation (22).

Let $V(s, a)$ denote the equilibrium value from state (s, a). The Bellman equation is

$$V(s,a) = ax_s + \max_{I \in \mathcal{I}(a)} \left\{ -k_s I + \beta \sum_{s' \in \mathcal{S}} P_{ss'} V\big(s', a + (1-a)I\big) \right\}. \tag{24}$$

Proposition 1 (Existence and uniqueness of the value)**.** *A stationary Markov perfect equilibrium exists. The equilibrium value function is unique, and at least one equilibrium is pure. The equilibrium strategy is unique whenever the two actions yield different values in both decision states. If the conventional sector is indifferent in either decision state, the value function remains unique and the equilibrium strategy is non-unique.*

The value of fiscal capacity We now turn to the long-run expected value of fiscal investment, comparing the utility path of making the investment to not making it. Write

$$W_s \equiv V(s,1), \qquad V_s \equiv V(s,0).$$

Installed capacity admits no further action. Define

$$D \equiv 1 - \beta(1-\pi) > 0, \qquad q \equiv \frac{\beta\pi}{D} \in (0,1). \tag{25}$$

The term q equals

$$q = \sum_{n=1}^{\infty} \beta^n \pi (1-\pi)^{n-1} = \mathbb{E}\big[\beta^\tau \mid s_0 = \ell\big]. \tag{26}$$

It is the expected discount factor applied to a payoff that begins when the AI transition occurs. The Markov process therefore affects the investment problem through a single sufficient statistic for the discounted timing of the transition.

Capacity produces X in every period after the transition and no current saving

before it. Its values in the two economic states are consequently

$$W_h = \frac{X}{1-\beta}, \tag{27}$$

$$W_\ell = qW_h = \frac{qX}{1-\beta}. \tag{28}$$

Investment cut-offs There are two ways in which investment can happen - before the transition, which we call anticipatory investment, and after the transition, which we call reactive investment. We begin by considering first the decision after the transition - hence reactive investment - and then returning to the initial state - anticipatory investment.

Beginning with reactive investment, suppose capacity is absent in state h. The conventional sector compares the political cost c/ρ with the discounted stream of savings that begins once capacity becomes available in the following period. Define the reactive investment cut-off

$$R(\rho) \equiv \frac{1-\beta}{\beta}\frac{c}{\rho}. \tag{29}$$

The conventional sector induces *reactive investment* after the transition if and only if the benefit they obtain from digital fiscal capacity in terms of reduced tax burden X is larger than the discounted political cost of securing investment $R(\rho)$.

$$X \geq R(\rho). \tag{30}$$

Now we return to state ℓ, to consider *anticipatory investment*. Investment costs c, produces no saving before the transition, and makes capacity available from the following period. If the conventional sector does not invest while the economy remains in state ℓ, stationarity implies that it waits until the transition. The value of waiting therefore depends on whether investment will be profitable in state h.

First suppose that reactive investment will be unprofitable. Waiting then leaves the economy without digital fiscal capacity after the transition. Define the anticipatory cut-off without subsequent reactive investment as

$$A_N \equiv \frac{(1-\beta)c}{q} = \frac{(1-\beta)D}{\beta\pi}c. \tag{31}$$

When $X < R(\rho)$, the conventional sector induces *anticipatory investment* before the transition if and only if

$$X \geq A_N. \tag{32}$$

This is not the only possible case of anticipatory investment, however. Suppose instead that reactive investment will be profitable. Waiting preserves the possibility of inducing investment after the transition, but conventional firms will then bear one period of blunt taxation and face the higher political cost c/ρ. Define the anticipatory cut-off with subsequent reactive investment as

$$A_R(\rho) \equiv \frac{c}{q} - \frac{c}{\rho} = \frac{D}{\beta\pi}c - \frac{c}{\rho}. \tag{33}$$

When $X \geq R(\rho)$, the conventional sector induces anticipatory investment before the transition if and only if

$$X \geq A_R(\rho). \tag{34}$$

Thus $R(\rho)$ governs reactive investment, A_N governs anticipatory investment when waiting leads to no investment after the transition, and $A_R(\rho)$ governs anticipatory investment when waiting preserves the possibility of a reactive response. Equality at any cut-off permits both actions.

Equilibrium regimes With these cut-offs in hand we are now in a position to look at which equilibriums prevail, given different parameters: no investment; anticipatory investment; or reactive investment. To do so, we define the critical level of post-transition relative political power as

$$\rho^* \equiv \frac{q}{\beta} = \frac{\pi}{D} = \frac{\pi}{1-\beta(1-\pi)} \in (0,1]. \tag{35}$$

This value determines whether reactive investment can arise along the equilibrium path and provides us with Proposition 2.

Proposition 2 (Equilibrium investment regimes)**.** *Suppose that X differs from the applicable investment cut-offs. From the initial state $(\ell, 0)$, the equilibrium path is characterised as follows.*

If $\rho > \rho^*$ (high future political power of conventional firms), then

$$R(\rho) < A_N < A_R(\rho), \tag{36}$$

and depending on the benefit of fiscal capacity to conventional firms X, we get:

$$\begin{cases} \textit{no investment}, & X < R(\rho), \\ \textit{reactive investment}, & R(\rho) < X < A_R(\rho), \\ \textit{anticipatory investment}, & X > A_R(\rho). \end{cases} \tag{37}$$

If $\rho < \rho^*$ (low future political power of conventional firms), then

$$A_R(\rho) < A_N < R(\rho), \tag{38}$$

and depending on the benefit of fiscal capacity to conventional firms X, we get:

$$\begin{cases} \textit{no investment}, & X < A_N, \\ \textit{anticipatory investment}, & X > A_N. \end{cases} \tag{39}$$

If $\rho = \rho^*$, then

$$R(\rho) = A_N = A_R(\rho).$$

Below their common value, there is no investment. Above the latter, investment is anticipatory.

Panel (a) of Figure 1 visualises Proposition 2. Along the x-axis we have the relative future political power of conventional firms ρ and along the y-axis we have the fiscal benefits of digital fiscal capacity to conventional firms X adjusted by the cost of political influence c: hence, X/c.

Each point $(\rho, X/c)$ identifies the equilibrium path resulting from the initial state. There are three possible equilibriums reached: No Investment; Anticipatory Investment; and Reactive Investment. The vertical positions of the labelled boundaries are $R(\rho)/c$, A_N/c, and $A_R(\rho)/c$. The figure displays only those cut-offs that delimit equilibrium paths in the relevant range of ρ.

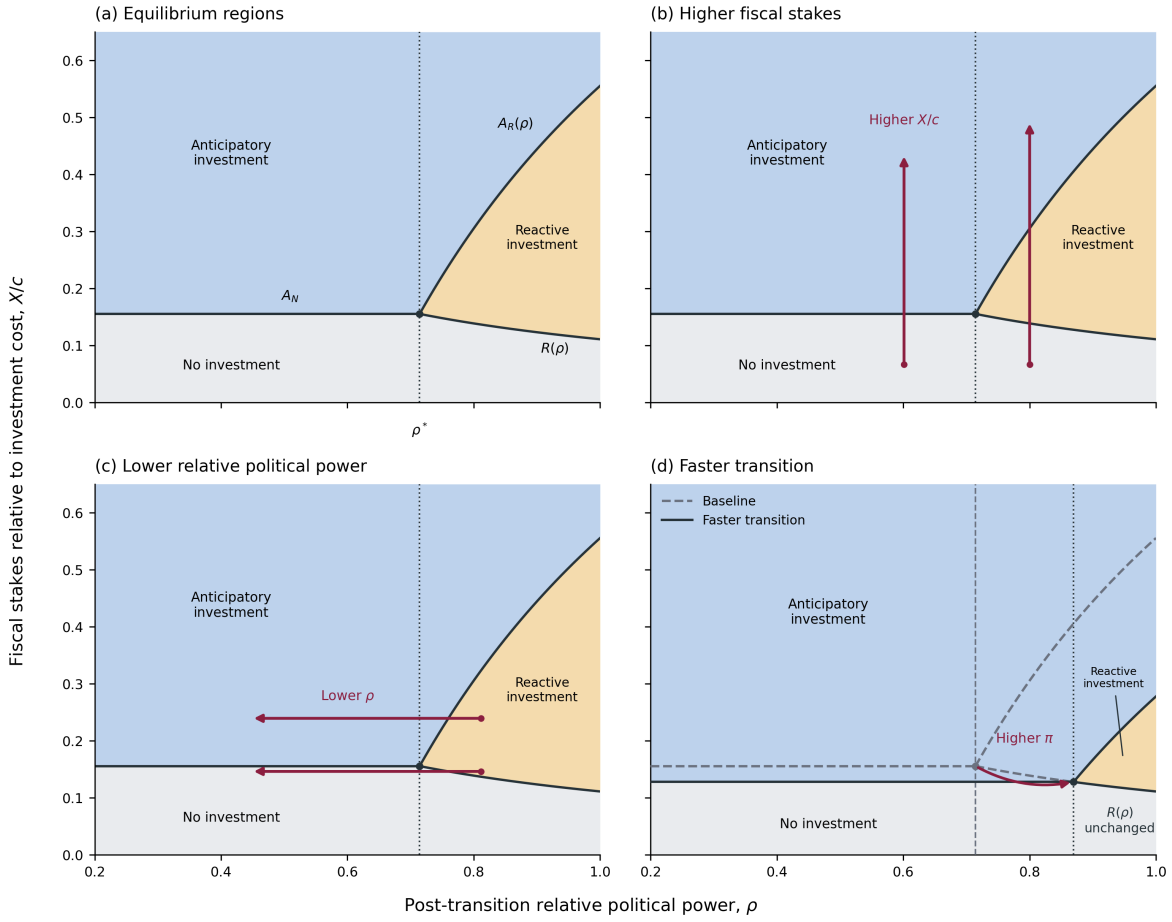


Figure 1: Equilibrium regimes and central comparative statics. Panel (a) shows the equilibrium partition. Panel (b) raises the normalised fiscal stake X/c, and panel (c) lowers post-transition relative political power ρ. Since these quantities appear on the axes, the arrows represent movements through a fixed partition. Panel (d) raises the transition probability π, with the dashed lines showing the initial boundaries and the solid lines the new boundaries.

To the left of ρ^*, the zone of low future political power for conventional firms, the line A_N separates no investment from anticipatory investment. To its right, the zone of higher future political power, the line $R(\rho)$ separates no investment from reactive investment, and $A_R(\rho)$ separates reactive from anticipatory investment.

$R(\rho)$ is the reactive cut-off because it determines whether conventional firms invest once state h has been reached. A_N is the anticipatory cut-off when waiting would leave capacity permanently absent because reactive investment is unprofitable. $A_R(\rho)$ is the anticipatory cut-off when waiting preserves the possibility of reactive investment. The three cut-offs coincide at ρ^*, where the option to wait ceases to create a distinct reactive region.

No investment means that fiscal capacity remains absent before and after the transition. Under reactive investment, conventional firms wait until state h, bear one period of blunt taxation, and then induce investment. Under anticipatory investment, they induce investment in state ℓ, so capacity is available before the transition can affect taxation.

The division around ρ^* has a political interpretation. When conventional firms retain sufficient relative political power after the transition, they can defer investment at intermediate fiscal stakes. Greater relative power reduces the subsequent cost of inducing investment, which lowers $R(\rho)$. It also makes waiting more attractive, which raises $A_R(\rho)$. The reactive region between the two cut-offs therefore widens as ρ increases. When post-transition relative power falls below ρ^*, this intermediate regime disappears. Conventional firms induce investment before the transition when $X > A_N$, including cases in which investment would be unprofitable afterwards. When $X < A_N$, the prospective saving is insufficient to justify the current cost and capacity remains absent. A sufficiently large prospective loss of relative political power can therefore produce either anticipatory investment or persistent administrative weakness, depending on the fiscal stakes.

At an investment cut-off, the conventional sector is indifferent between its available actions (see the appendix for the corresponding stationary strategies, off-path actions, and equality cases).

2.3 Comparative statics

We now turn to examining how various parameters in the model affect the likely equilibrium investment. The central comparative statics concern (i) the fiscal stake

created by the AI transition; (ii) the accompanying change in the relative political power of conventional firms; and (iii) the speed with which the transition occurs. Panels (b)–(d) of Figure 1 show how each change affects the equilibrium map. Since X/c and ρ appear on the axes, changes in these quantities move an economy vertically or horizontally through the partition shown in panel (a). The transition probability π does not appear on either axis, so changing it shifts the boundaries themselves. Appendix B.5 provides the derivations.

Proposition 3 (Central comparative statics)**.** *Suppose that the initial and final parameter values lie away from the cut-off equalities.*

1. Fiscal stakes. *Conditional on β, π, and ρ, the quantities S, i_h, χ, and c affect equilibrium investment through the normalised stake*

$$\frac{X}{c} = \frac{\chi\alpha(i_h)S}{c}. \tag{40}$$

This stake is strictly increasing in S and χ, weakly increasing in i_h, and strictly decreasing in c. An increase in X/c weakly raises the stationary investment action in each state without capacity. Along the equilibrium path, an increase in X/c produces a movement from no investment to reactive investment and then to anticipatory investment when $\rho > \rho^*$. When $\rho < \rho^*$, the path moves directly from no investment to anticipatory investment.

2. Relative political power. *Conditional on X, c, π, and β, an increase in ρ satisfies*

$$\frac{\partial R(\rho)}{\partial \rho} < 0, \qquad \frac{\partial A_R(\rho)}{\partial \rho} > 0, \qquad \frac{\partial A_N}{\partial \rho} = 0. \tag{41}$$

It weakly raises the investment action in state $(h, 0)$ and weakly lowers the investment action in state $(\ell, 0)$. Equivalently, a prospective loss of relative political power weakly encourages investment before the transition and weakly discourages investment after it. If $X > A_N$, a fall in ρ can replace reactive investment with anticipatory investment. If $X < A_N$, it can replace reactive investment with no investment.

3. Transition speed. *Conditional on X, c, ρ, and β, an increase in π satisfies*

$$\frac{\partial R(\rho)}{\partial \pi} = 0, \qquad \frac{\partial A_N}{\partial \pi} < 0, \qquad \frac{\partial A_R(\rho)}{\partial \pi} < 0, \qquad \frac{\partial \rho^*}{\partial \pi} > 0. \tag{42}$$

It leaves the investment action in state $(h, 0)$ unchanged and weakly raises the investment action in state $(\ell, 0)$. The possible changes in the equilibrium path are from no investment to anticipatory investment and from reactive investment to anticipatory investment.

Proof. See Appendix B.5. □

We now discuss each of these comparative statics at more length:

Fiscal stakes Increasing, the fiscal benefit of digital fiscal capacity to conventional firms, unsurprisingly makes investment of any kind more likely. Panel (b) of Figure 1 represents an increase in X/c as an upward movement. The cut-offs remain unchanged. When future conventional firm political power is high—to the right of ρ^*—the upward arrow crosses $R(\rho)/c$ and then $A_R(\rho)/c$, producing the sequence from no investment to reactive investment and then to anticipatory investment. Where, future political power is low—to the left of ρ^*—it crosses only A_N/c, so the path moves directly from no investment to anticipatory investment.

Equation (40) collects the fiscal shortfall S, inequality i_h, the incidence wedge χ, and the cost of investment c into one sufficient statistic. Accordingly we can examine the effects of these subcomponents on the equilibrium choice. A larger fiscal shortfall S increases the revenue that must be obtained from capital. Greater inequality raises the share $\alpha(i_h)$ assigned to capital, since $\alpha'(i_h) > 0$. A larger incidence wedge χ increases the share of this contribution that digital fiscal capacity can transfer from conventional firms to AI firms. An increase in any one of these forces makes investment more likely. A larger investment cost c, unsurprisingly works in the opposite direction.

We can also examine interactions between these components. The first three components enter multiplicatively. In particular,

$$\frac{\partial^2 (X/c)}{\partial S \, \partial i_h} = \frac{\chi}{c} \alpha'(i_h) \geq 0. \tag{43}$$

Inequality therefore matters more when erosion of the existing tax base is large, and a given fiscal shortfall matters more when electoral pressure assigns a large share of it to capital. An increase in inequality creates no private benefit from capacity in the absence of a fiscal requirement. Likewise, erosion of the existing tax base produces a weaker investment incentive when the politically required capital share is small.

In other words, inequality can indeed produce greater digital fiscal capacity but it has to run through both fiscal pressures and a wedge between conventional and AI firms in how they are affected by blunt versus targeted taxation. In the absence of either, we should not expect an effect of inequality.

Relative political power Panel (c) represents a prospective loss of relative political power as a movement to the left. The curves remain fixed because their slopes already record how the cut-offs vary with ρ. Moving left raises $R(\rho)/c$, making reactive investment more difficult, and lowers $A_R(\rho)/c$, making anticipatory investment more attractive. The horizontal boundary A_N/c is independent of ρ.

The two arrows in panel (c) show why the effect depends on the fiscal stakes. When $X > A_N$, a sufficiently large fall in ρ replaces reactive investment with anticipatory investment. Conventional firms establish capacity before their relative power deteriorates. When $X < A_N$, the current benefit is insufficient to justify anticipatory investment, while the loss of power can make subsequent reactive investment unprofitable. The path then moves from reactive investment to no investment. The same prospective political displacement can therefore induce early investment or leave the state persistently without targeted capacity, depending on the relevant fiscal stakes to conventional firms.

The speed of the transition Panel (d) compares the boundaries under the initial transition probability, shown by the dashed lines, with those under a higher value of π, shown by the solid lines. A faster transition brings the expected benefits from capacity closer to the present and increases the expected cost of waiting in state ℓ. It therefore shifts both A_N/c and $A_R(\rho)/c$ downwards. It also shifts ρ^* to the right, so conventional firms must retain more relative power for a reactive region to exist. The reactive region consequently contracts.

The reactive cut-off $R(\rho)/c$ does not move. Once state h has been reached, the previous speed of the transition no longer affects the value of investment. Under the timing assumptions of the baseline, a faster transition therefore weakly favours earlier investment.[11]

[11]This monotonic result depends on the assumption that investment chosen in state ℓ survives the transition and produces capacity in the following period. A transition that could interrupt an unfinished administrative reform, prevent its completion, or remove the authority required to use it could generate an opposing effect. Such an extension would be required for the model to produce a transition that occurs too quickly for capacity to be established.

Overall, the model incorporates the rise of artificial intelligence in three distinct ways. It distinguishes the magnitude of the fiscal stakes (including the collapse in the labour share, higher inequality and the relative taxability of conventional versus AI firms), from the speed of the AU transition, and from the accompanying redistribution of political power between AI and conventional firms following the transition. High fiscal stakes make capacity valuable, a faster transition increases the cost of delay, and a large prospective loss of power reduces the feasibility of investment after the transition.

The interaction between fiscal stakes and relative political power produces the central ambiguity of the model. Anticipated political weakness can induce conventional firms to secure capacity in advance, or it can prevent investment after the transition without making earlier investment worthwhile. A scalar measure of AI exposure that merged all these considerations would conceal these important distinctions.

3 Discussion

The central result of the model—that an anticipated loss of relative power can produce anticipatory investment at high fiscal stakes and persistent low capacity at lower fiscal stakes—relies on two key assumptions. Shifting the tax burden from conventional firms towards AI rents requires specialised fiscal capacity, and capacity built up before AI firms gain relative power continues to affect taxation afterwards. Both assumptions can be reasonably called into question. For one, governments may be able to tax the AI sector without having to invest in digital fiscal capacity. And, allowing for limited commitment and thus commitment problems to arise means that, even when capacity is built up early, it may fail to secure future tax savings, especially if relative power changes sufficiently. These observations raise the question as to whether our mechanism is robust to relaxing these assumptions, which we examine next.[12]

Energy taxation and political power. Energy taxation provides perhaps the strongest version of what one might call the substitution objection to our argument. Energy

[12]Our tentative position on the other assumptions is that they are either fairly innocuous or would not substantially change the logic of the model while complicating the formal analysis. Given that this work in progress, however, we may well be wrong.

is one of the "means of prediction" (Kasy, 2025), and its physical use is easier to observe than rents derived from data, models, and organisational capital. For instance, a levy on electricity used for AI computing could raise revenue without requiring the government to distinguish rents from normal returns. Identifying the relevant energy use would itself require administrative rules, though this may demand less information than assessing the returns generated by AI.

In the model, such a levy or tax would reduce the residual fiscal requirement. If it also reduced AI rents or slowed the sector's expansion, it could preserve the relative power of conventional firms and delay the transition. Each effect weakens the incentive for anticipatory investment. Energy taxation could therefore allow governments to raise revenue from the AI sector while reducing the pressure to develop more precise methods for taxing its rents.

Allowing policymakers to resort to energy taxes changes the central argument only when the observable tax base provides a sufficiently "good" substitute for digital fiscal capacity. The levy must raise substantial revenue, be difficult to avoid through relocation or substitution, and fall mainly on AI rents rather than households or conventional firms. Under these conditions, the residual fiscal requirement and the private value of specialised capacity may become too small to induce conventional firms to support investment. If the tax raises only part of the required revenue or is largely passed through, the original mechanism remains, though investment becomes less likely. Land taxation provides a useful comparison. Its base is observable and immobile, meaning it can reduce the residual fiscal requirement, but it has little direct effect on the expansion or relative power of the AI sector unless it captures land rents generated specifically by AI production.[13]

A related concern is whether AI firms' rents are directly a function of energy use. To provide a relevant example, there is an ongoing debate among economists about various analogues of "robot taxes" that might prevent an AI transition from

[13]Voter responses would depend on incidence and revenue use. A general electricity tax resembles consumption taxation and may impose a larger burden relative to income on poorer households. It could therefore strengthen demands for capital to bear a greater share of the fiscal adjustment. A levy confined to energy used for AI computing would be more attractive when firms absorb it and the revenue finances transfers, lower labour taxation, or infrastructure costs created by data centres. Pass-through into electricity prices, AI services, or other goods would weaken support. Large incumbents may also be better able to relocate, obtain cheaper energy, or absorb the levy than smaller competitors. Energy taxation could then reduce the relative power of the AI sector while increasing concentration in the latter.

entirely replacing human labour by essentially taxing its replacement.[14] The merits of robot taxes depend on whether humans can easily move to non-routinisable work (Guerreiro et al., 2022) and whether the productivity gains of replacing humans with robots—or AI agents—are large enough to fully compensate displaced humans (Korinek and Lockwood, 2026). But it is not entirely obvious that displacing humans, or less efficient AI, will mean equivalent energy usage. Indeed, well-designed algorithms might mean overall energy usage goes down, even as rents to AI firms go up. This after all, would represent higher productivity.

Irwin et al. (2026) discuss a variety of options, including standard robot taxes that estimate job displacement, through direct capital gains taxation, and so-called FLOPS taxes (taxes on floating points / compute used) and token taxes (taxes on tokens produced as AI output). They argue that FLOPS and token taxes have the benefit of being easier to monitor than, say, direct job displacement, but they clearly both require substantial fiscal capacity to develop, along the lines suggested in this paper. An interesting difference in taking FLOPS/compute versus tokens is that with more efficient processes the same tokens—and hence same profitable output—may be produced with much lower levels of compute. Hence an energy tax, which picks up inputs like compute but not outputs like tokens, may fail to effectively tax AI firms' profits, thereby widening the gap between the state's ability to tax conventional versus AI firms. Further research into which types of fiscal capacity best make legible underlying AI profits would help distinguish which types of 'AI taxation' might heighten or weaken the underlying political competition over digital fiscal capacity outlined in our model.

Persistence and political commitment. The second point concerns the implications of relaxing the perfect commitment assumption. The absorbing high economic state requires the fiscal shortfall, inequality, and the new distribution of relative power to be persistent. The law of motion for capacity makes it durable once it has been built up. The incidence equations further assume that future policymakers continue to use that capacity, i.e. that commitment is perfect.

Note, first that relaxing the absorbingness of the high state has a separate effect

[14] Related proposals were already discussed across Western Europe in the late 1970s and 1980s. For instance, Herbert Ehrenberg (SPD) in West Germany and Alfred Dallinger (SPÖ) in Austria proposed basing employers' contributions to social insurance partly on value added rather than wages, while Lord Fairfax of Cameron, a Conservative peer, later suggested taxing the owners of automated production systems in the United Kingdom.

from relaxing commitment. If the economic transition can reverse, the expected duration of the fiscal savings falls and investment becomes less attractive. This generally raises the fiscal stakes required for investment without changing the reason conventional firms might prefer to act before losing relative power. A commitment problem is more consequential because it can remove the protection provided by early investment.

Consider first a limited commitment problem that is unrelated to the distribution of political power. Suppose that capacity is used with some constant probability and that this probability is the same whether capacity was built before or after the transition. Imperfect commitment then reduces the expected benefit from capacity and raises the fiscal stakes required for both anticipatory and reactive investment. It expands the region in which no investment occurs without changing the ordering of the regimes or the critical level of relative power separating them.[15]

The central comparative static can change when future use depends on the post-transition distribution of relative power. A decline in conventional firms' relative power then has two opposing effects. It raises the cost of building capacity after the transition, which favours anticipatory investment, while reducing the expected use of capacity built beforehand, which discourages it. The argument survives when the first effect dominates. It weakens or reverses when the loss of political commitment is sufficiently large.[16] In the limiting case in which AI firms can induce future policymakers to leave capacity already built up unused, early investment provides no protection. The anticipatory result therefore requires legal authority, reporting obligations, accumulated information, international agreements, or specialist bureaucracies that make capacity built under an earlier distribution of political power difficult for later governments to reverse. Put differently, our argument requires a

[15]Let $\lambda \in (0,1]$ denote the probability that available capacity is used in any high-state period. Replacing X with λX gives

$$R_\lambda(\rho) = \frac{R(\rho)}{\lambda}, \qquad A_{N,\lambda} = \frac{A_N}{\lambda}, \qquad A_{R,\lambda}(\rho) = \frac{A_R(\rho)}{\lambda}.$$

Since all three cut-offs are scaled by the same factor, the critical value ρ^* remains unchanged.

[16]Suppose that the probability of use is $\lambda(\rho)$, with $\lambda'(\rho) > 0$. In the region where reactive investment is otherwise possible, a decline in ρ continues to favour anticipatory investment when

$$\frac{\lambda'(\rho)}{\lambda(\rho)} < \frac{q}{\rho(\rho - q)}.$$

Reversing this inequality reverses the comparative static.

sufficient degree of persistence, but not perfect commitment, which is perhaps the most important scope condition of our argument.

4 Conclusion

Our starting point was Schumpeter's claim that changes in economic organisation can produce "a crisis of the old fiscal methods" and ultimately the tax state itself (Schumpeter, 1991, 101). The rise of artificial intelligence presents a contemporary version of this problem. However, much of the existing literature on the political economy of AI reduces the latter to one simple parameter (perhaps in growth or inequality). By contrast, our model argues that the rise of AI should affect our understanding of fiscal policy and fiscal capacity in five distinct ways that may reinforce each other.

First, and most discussed in the literature, it may reduce revenue from labour taxation forcing government to consider other forms of taxation, typically on consumption and on capital. Second, if AI increases inequality, it is likely to create greater electoral demand for capital to take the greater burden of increased taxation. Third, AI generates rents that governments struggle to identify and value when taxing capital, creating a tension among capital owners—with conventional firms more exposed to blunt taxation on identifiable assets than AI firms. This potentially creates a demand for the building of digital fiscal capacity. Fourth, the speed of the AI transition affects the timing of the creation of this digital fiscal capacity—if the move to artificial general intelligence (AGI) is likely to soon and sudden, conventional firms may lobby hard for anticipatory fiscal investment. Finally, AI alters the relative political balance of power between conventional firms and AI firms post the transition to an AI economy. If conventional firms foresee a collapse in their power, they may be incentivised to move now or forever be subject to blunt taxation.

Our model shows that declining revenue from labour taxation and greater inequality strengthen conventional firms' support for digital fiscal capacity. Fiscal stakes, the speed of the transition, and their political power relative to AI firms determine whether and when investment occurs. A faster transition encourages anticipatory investment. An anticipated loss of influence can induce early investment when fiscal stakes are high or prevent later investment when they are low. The

growing power of AI firms can therefore accelerate fiscal innovation or leave the state reliant on old fiscal methods. There is an indeterminacy in whether the fiscal state will be able to keep up with the rise of AI.

Our argument locates investment in fiscal capacity in intra-business conflict over tax incidence. Firms disadvantaged by established tax instruments may support institutions that extend taxation to an ascendant sector. The mechanism requires specialised capacity for targeted taxation and sufficient durability of capacity established before the transition. Substitutes, such as energy taxation, can reduce demand for digital fiscal capacity. Political obstruction after the transition can reduce the value of capacity established in advance.

To return to where we began, the resolution of Schumpeter's crisis in an era of AI depends on whether firms burdened by old fiscal methods can induce investment before AI firms acquire sufficient power to block it. As Acemoglu and Restrepo (2017) famously argued, there is indeed a race between man and machine as the latter substitutes for the former. But there is also a new race between firms that traditionally relied on humans and those whose profits come from a brave new world of hyper-intelligent AI agents. How that political battle resolves will determine the future of the fiscal state.

References

Acemoglu, Daron, "The simple macroeconomics of AI," *Economic Policy*, January 2025, *40* (121), 13–58.

_ , A Arda Gitmez, and Mehdi Shadmehr, "Automation and Repression," *Working Paper*, 2026.

_ and Pascual Restrepo, "The Race between Machine and Man: Implications of Technology for Growth, Factor Shares and Employment," *The American Economic Review*, June 2017.

_ , Asuman Ozdaglar, and James Siderius, "AI and Social Media: A Political Economy Perspective," in Ajay Agrawal, Joshua Gans, Avi Goldfarb, and Catherine E. Tucker, eds., *The Political Economy of Artificial Intelligence*, Chicago: University of Chicago Press, 2026, p. Chapter 12.

Aghion, Philippe and Simon Bunel, "AI and Growth: Where Do We Stand?," *Policy Note*, 2024.

Andersson, Per F., "Taxation and Left-Wing Redistribution: The Politics of Consumption Tax in Britain and Sweden," *Comparative Politics*, January 2022, *54* (2), 279–301.

Ansell, Ben and David Samuels, "Inequality and Democratization: A Contractarian Approach," *Comparative Political Studies*, December 2010, *43* (12), 1543–1574.

_ **and** _ , *Inequality and Democratization: An Elite-Competition Approach*, New York, NY: Cambridge University Press, February 2015.

Atkinson, Anthony B., *Inequality: What Can Be Done?*, Cambridge, Massachusetts: Harvard University Press, 2015.

Beraja, Martin, Andrew Kao, David Y Yang, and Noam Yuchtman, "AI-tocracy," *The Quarterly Journal of Economics*, August 2023, *138* (3), 1349–1402.

_ **, David Y Yang, and Noam Yuchtman**, "Data-intensive Innovation and the State: Evidence from AI Firms in China," *The Review of Economic Studies*, July 2023, *90* (4), 1701–1723.

Beramendi, Pablo and David Rueda, "Social Democracy Constrained: Indirect Taxation in Industrialized Democracies," *British Journal of Political Science*, October 2007, *37* (4), 619–641.

_ **, Mark Dincecco, and Melissa Ziegler Rogers**, "Intra-Elite Competition and Long-Run Fiscal Development," *The Journal of Politics*, September 2016.

Besley, Timothy and Torsten Persson, "The Origins of State Capacity: Property Rights, Taxation, and Politics," *American Economic Review*, September 2009, *99* (4), 1218–1244.

_ **and** _ , *Pillars of Prosperity: The Political Economics of Development Clusters*, Princeton, NJ: Princeton University Press, August 2011.

Boix, Carles, "AI and the Economic and Informational Foundations of Democracy," in Justin B. Bullock, Yu-Che Chen, Johannes Himmelreich, Valerie M. Hudson, Anton Korinek, Matthew M. Young, and Baobao Zhang, eds., *The Oxford Handbook of AI Governance*, Oxford: Oxford University Press, April 2024, pp. 707–725.

Brambor, Thomas, Agustín Goenaga, Johannes Lindvall, and Jan Teorell, "The Lay of the Land: Information Capacity and the Modern State," *Comparative Political Studies*, February 2020, *53* (2), 175–213.

Brynjolfsson, Erik, Daniel Rock, and Chad Syverson, "The Productivity J-Curve: How Intangibles Complement General Purpose Technologies," *American Economic Journal: Macroeconomics*, January 2021, *13* (1), 333–372.

Callander, Steven, "Regulating AI: The Race Between Policymakers and the Rise of a New Interest Group," in Ajay Agrawal, Joshua Gans, Avi Goldfarb, and Catherine E. Tucker, eds., *The Political Economy of Artificial Intelligence*, Chicago, IL: University of Chicago Press, 2026, p. Chapter 3.

Cusack, Thomas R. and Pablo Beramendi, "Taxing work," *European Journal of Political Research*, 2006, *45* (1), 43–73.

Farboodi, Maryam and Laura Veldkamp, "Data and Markets," *Annual Review of Economics*, September 2023, *15* (Volume 15, 2023), 23–40.

_ and _, "A Model of the Data Economy," *The Review of Economic Studies*, July 2026, *Forthcoming*, rdag074.

Fearon, James D, "Rationalist explanations for war," *International organization*, 1995, *49* (3), 379–414.

Fetzer, Thiemo, Callum Shaw, and Jacob Edenhofer, "Informational Boundaries of the State," *CESifo Working Paper Series*, 2024. Number: 10901.

Furman, Jason and Robert Seamans, "AI and the Economy," *Innovation Policy and the Economy*, January 2019, *19*, 161–191.

Garfias, Francisco, "Elite Competition and State Capacity Development: Theory and Evidence from Post-Revolutionary Mexico," *American Political Science Review*, May 2018, *112* (2), 339–357.

_ and Emily A. Sellars, "Fiscal legibility and state development: Theory and evidence from colonial Mexico," *American Journal of Political Science*, 2026, *70* (1), 5–20.

Grant, Zack, Jane Green, and Geoffrey Evans, "New Tech, New Threat? Occupational Exposure to Artificial Intelligence Increases Support for AI Regulation," December 2025.

Gratton, Gabriele and Barton E. Lee, "Market Power Is Power," *SSRN Working Paper*, May 2026.

Guerreiro, Joao, Sergio Rebelo, and Pedro Teles, "Should robots be taxed?," *The Review of Economic Studies*, 2022, *89* (1), 279–311.

Haskel, Jonathan and Stian Westlake, *Capitalism without Capital: The Rise of the Intangible Economy*, Princeton: Princeton University Press, 2018.

Haslberger, Matthias, Jane Gingrich, and Jasmine Bhatia, "Rage against the machine? Generative AI exposure, subjective risk, and policy preferences," *Journal of European Public Policy*, September 2025, *0* (0), 1–28.

Healy, Patrick, Simon D. Angus, Paul Raschky, Klaus Ackermann, Nathan Lane, Weijia Li, and Cynthia A. Huang, "Digital State Capacity," July 2026.

Irwin, Lucas, Tung-Yu Wu, and Fazl Barez, "Token Taxes: mitigating AGI's economic risks," *arXiv preprint arXiv:2603.04555*, 2026.

Jones, Charles I., "A.I. and Our Economic Future," *Journal of Economic Perspectives*, 2026.

_ **and Christopher Tonetti**, "Nonrivalry and the Economics of Data," *American Economic Review*, September 2020, *110* (9), 2819–2858.

Kasy, Maximilian, *The Means of Prediction: How AI Really Works*, Chicago: University of Chicago Press, 2025.

_ , "The means of prediction and the production function of AI," *Journal of Economic Interaction and Coordination*, August 2026.

Keen, Michael and Joel Slemrod, *Rebellion, Rascals, and Revenue: Tax Follies and Wisdom through the Ages*, Princeton; Oxford: Princeton University Press, 2022.

Korinek, Anton and Lee Lockwood, "Public Finance in the Age of AI: A Primer," *NBER Working Paper*, February 2026, *w34873.*

Kurz, Mordecai, *The Market Power of Technology: Understanding the Second Gilded Age*, New York: Columbia University Press, 2023.

Lall, Ranjit, "The Political Economy of Artificial Intelligence: Evidence from Western Europe," August 2025.

Lee, Melissa M. and Nan Zhang, "Legibility and the Informational Foundations of State Capacity," *The Journal of Politics*, January 2017, *79* (1), 118–132.

Li, Weijia, Gérard Roland, and Yang Xie, "Erosion of State Power, Corruption Control and Fiscal Capacity," *The Economic Journal*, May 2022, *132* (644), 1542–1565.

Magistro, Beatrice, Sophie Borwein, R. Michael Alvarez, Bart Bonikowski, and Peter John Loewen, "Attitudes toward artificial intelligence (AI) and globalization: Common microfoundations and political implications," *American Journal of Political Science*, 2026, *Forthcoming.*

Mansell, Jordan, Alexander Beyer, Ori Frieman, John McAndrews, and Clifton van der Linden, "Political Preferences and Support for Artificial Intelligence," in Ajay Agrawal, Joshua Gans, Avi Goldfarb, and Catherine E. Tucker, eds., *The Political Economy of Artificial Intelligence*, University of Chicago Press, 2026, p. Chapter 11.

Mares, Isabela and Didac Queralt, "Fiscal innovation in nondemocratic regimes: Elites and the adoption of the prussian income taxes of the 1890s," *Explorations in Economic History*, July 2020, *77*, 101340.

Meltzer, Allan H. and Scott F. Richard, "A Rational Theory of the Size of Government," *Journal of Political Economy*, 1981, *89* (5), 914–927.

Ranaldi, Marco, "Broadening Capital Ownership in the Age of AI," August 2026.

Restrepo, Pascual, "We Won't be Missed: Work and Growth in the AGI World," October 2025.

Schumpeter, Joseph, *Joseph A. Schumpeter: The Economics and Sociology of Capitalism*, Princeton, N.J.: Princeton University Press, 1991.

Scott, James C., *Seeing Like a State: How Certain Schemes to Improve the Human Condition Have Failed*, New Haven: Yale University Press, February 1999.

Thewissen, Stefan and David Rueda, "Automation and the Welfare State: Technological Change as a Determinant of Redistribution Preferences," *Comparative Political Studies*, February 2019, *52* (2), 171–208.

Veldkamp, Laura, "Valuing Data as an Asset," *Review of Finance*, September 2023, *27* (5), 1545–1562.

Veldkamp, Laura L. and Isaac Baley, *The Data Economy: Tools and Applications*, Princeton; Oxford: Princeton University Press, 2025.

Yang, Eddie, "The limits of AI for authoritarian control," *American Journal of Political Science*, 2026, *Forthcoming.*

Appendix

A Mobilisation technology

The baseline represents the conventional sector as choosing whether to induce investment and, if it does so, paying the political cost k_s. This subsection derives that binary representation from a continuous choice of mobilisation effort.

Suppose that capacity is absent. The conventional sector chooses mobilisation effort $m_t \geq 0$, measured in payoff units, and incurs the current cost m_t. Let $\varrho_s > 0$ denote the political effectiveness of this effort in economic state s, and let $\bar{c} > 0$ denote the amount of effective political support required to induce investment. The induced investment outcome is

$$I(m; s, a) = (1 - a)\mathbf{1}\left\{\varrho_s m \geq \bar{c}\right\}. \tag{44}$$

When capacity is already available, further mobilisation is neither necessary nor feasible. The weak inequality in equation (44) means that effort at the requirement succeeds. Investment induced in period t enters the capacity law in equation (16) and therefore first affects taxation in period $t + 1$.

Any effort

$$m \in \left(0, \frac{\bar{c}}{\varrho_s}\right)$$

is strictly dominated by zero effort because it incurs a positive cost without inducing investment. Any effort

$$m > \frac{\bar{c}}{\varrho_s}$$

is strictly dominated by $\bar{c}/\varrho_s$ because both induce investment while the latter costs less. The only undominated choices in state $(s, 0)$ are consequently

$$m \in \left\{0, \frac{\bar{c}}{\varrho_s}\right\}. \tag{45}$$

Define the effective political cost of inducing investment as

$$k_s \equiv \frac{\bar{c}}{\varrho_s}. \tag{46}$$

On the undominated action set, mobilisation expenditure is exactly

$$m = k_s I.$$

Removing the dominated effort choices therefore yields the binary action and the payoff term $-k_s I_t$ used in the main model.

Define the pre-transition cost and the change in political effectiveness as

$$c \equiv \frac{\bar{c}}{\varrho_\ell}, \qquad \rho \equiv \frac{\varrho_h}{\varrho_\ell} \in (0, 1]. \tag{47}$$

It follows that

$$k_\ell = c, \qquad k_h = \frac{c}{\rho}, \tag{48}$$

which gives equation (17). The restriction $\rho \leq 1$ imposes a weak decline in the political effectiveness of conventional firms after the transition. Since

$$\rho = \frac{k_\ell}{k_h},$$

a lower value means that inducing investment requires more political resources after the transition.

The relative-power interpretation can be made explicit by allowing for opposition from AI firms. Suppose that investment is induced when

$$\theta_s^N m_t \geq b + \theta_s^A \bar{m}_{A,s}, \tag{49}$$

where $b > 0$ is an institutional hurdle, $\bar{m}_{A,s} \geq 0$ is the opposition mounted by AI firms in state s, and $\theta_s^N > 0$ and $\theta_s^A \geq 0$ measure the political effectiveness of the two sectors. Treating $\bar{m}_{A,s}$ as fixed within each state preserves the single-actor decision problem in the baseline. The minimum successful effort by conventional firms is

$$\kappa_s \equiv \frac{b + \theta_s^A \bar{m}_{A,s}}{\theta_s^N}. \tag{50}$$

Setting

$$c = \kappa_\ell \qquad \text{and} \qquad \rho = \frac{\kappa_\ell}{\kappa_h}$$

reproduces equation (48). Equivalently,

$$\rho = \frac{\dfrac{\theta_h^N}{b + \theta_h^A \bar{m}_{A,h}}}{\dfrac{\theta_\ell^N}{b + \theta_\ell^A \bar{m}_{A,\ell}}}. \tag{51}$$

A decline in the political effectiveness of conventional-sector mobilisation, an increase in the resources or effectiveness of AI-sector opposition, or both, lowers ρ. Endogenising mobilisation by AI firms would instead produce a political contest and would require their payoffs and costs to enter the dynamic game.

The timing follows the baseline. Mobilisation in period t uses ϱ_{s_t}, and its cost is paid immediately. Taxes in that period depend on a_t, while induced investment changes a_{t+1}. Anticipatory mobilisation therefore costs c, including when the transition occurs at the end of the same period, and capacity is available in the first period after the transition. Reactive mobilisation costs c/ρ. Conventional firms bear blunt capital taxation in the period of mobilisation and receive the saving from capacity from the following period onwards. At an investment cut-off, zero effort and the minimum successful effort are both optimal, corresponding to the mixed stationary strategies described in Appendix B.4.

The exact inverse relation between political power and cost follows from the fixed requirement and linear mobilisation cost. With a strictly increasing cost function $g(m)$, where $g(0) = 0$, the effective cost would be

$$g\left(\frac{\bar{c}}{\varrho_s}\right).$$

The binary reduction would remain valid, although the post-transition cost could no longer generally be written as c/ρ.

B Proofs and additional derivations

B.1 Proof of Proposition 1

Proof. Let $\mathcal{T}$ denote the operator on the right-hand side of equation (24). Current payoffs are bounded, the state space is finite, and every action set is finite. For any

bounded value functions V and $\widetilde{V}$,

$$\left\|\mathcal{T}V - \mathcal{T}\widetilde{V}\right\|_\infty \leq \beta \left\|V - \widetilde{V}\right\|_\infty.$$

The operator is a contraction because $\beta \in (0,1)$. It therefore has a unique fixed point. A maximising action exists at every state, and any stationary selection from the maximising actions is optimal at every state. A pure selection always exists. Strict maximisation gives a unique action. An equality permits every mixture over the maximising actions and leaves the fixed point unchanged. □

B.2 Value functions and investment conditions

When capacity exists in state h, the Bellman equation gives

$$W_h = X + \beta W_h = \frac{X}{1-\beta}.$$

When capacity exists in state ℓ, it produces no current saving and remains available after either transition. Hence

$$W_\ell = \beta\left[(1-\pi)W_\ell + \pi W_h\right] = \frac{\beta\pi}{D}W_h = qW_h.$$

When capacity is absent in state h, the conventional sector can wait or invest. Its value satisfies

$$V_h = \max\left\{\beta V_h, \quad -\frac{c}{\rho} + \beta W_h\right\}. \tag{52}$$

Waiting in every subsequent period gives zero. Equation (52) therefore becomes

$$V_h = \max\left\{0, \quad -\frac{c}{\rho} + \frac{\beta X}{1-\beta}\right\}. \tag{53}$$

Investment is optimal precisely when $X \geq R(\rho)$.

When capacity is absent in state ℓ, immediate investment has value $-c + W_\ell$. Waiting preserves absent capacity while the economic state evolves. The Bellman equation is

$$V_\ell = \max\left\{-c + W_\ell, \quad \beta\left[(1-\pi)V_\ell + \pi V_h\right]\right\}. \tag{54}$$

If the conventional sector waits whenever state ℓ recurs, its value is qV_h. Thus

$$V_\ell = \max\{-c + W_\ell, \quad qV_h\}. \tag{55}$$

Stationarity implies that inaction in state ℓ continues until the economic state changes. The strategy cannot prescribe investment after a fixed number of further periods in state ℓ because the payoff-relevant state remains unchanged.

If investment in state h is unprofitable, then $V_h = 0$. Immediate investment in state ℓ is optimal precisely when

$$-c + W_\ell \geq 0,$$

which is equivalent to $X \geq P$. If investment in state h is profitable, equation (53) gives

$$V_h = -\frac{c}{\rho} + \frac{\beta X}{1-\beta}.$$

Substitution into equation (55) shows that immediate investment in state ℓ is optimal precisely when $X \geq A(\rho)$.

B.3 Proof of Proposition 2

Proof. Equation (53) gives $I_h^* = 1$ precisely when $X > R(\rho)$. If $X < R(\rho)$, equation (55) gives $I_\ell^* = 1$ precisely when $X > P$. If $X > R(\rho)$, the same equation gives $I_\ell^* = 1$ precisely when $X > A(\rho)$.

The differences between the cut-offs are

$$P - R(\rho) = \frac{(1-\beta)c}{\beta}\left(\frac{D}{\pi} - \frac{1}{\rho}\right), \tag{56}$$

$$A(\rho) - P = c\left(\frac{D}{\pi} - \frac{1}{\rho}\right), \tag{57}$$

$$A(\rho) - R(\rho) = \frac{c}{\beta}\left(\frac{D}{\pi} - \frac{1}{\rho}\right). \tag{58}$$

All three differences have the sign of $\rho - \rho^*$. This establishes equations (36) and (38). Combining each ordering with the investment conditions in the two states yields equations (**??**) and (**??**). At $\rho = \rho^*$, the differences are zero and the cut-offs coincide. □

B.4 Non-uniqueness at indifference

The stationary equilibrium strategy is unique away from the cut-off equalities. If $\rho > \rho^*$, the action in state h is non-unique at $X = R(\rho)$, and the action in state ℓ is non-unique at $X = A(\rho)$. If $\rho < \rho^*$, the action in state ℓ is non-unique at $X = P$, and the action in state h is non-unique at $X = R(\rho)$. If $\rho = \rho^*$ and $X = R(\rho) = P = A(\rho)$, the equilibrium action is non-unique in both decision states.

Equations (53) and (55) establish these statements directly. Two actions maximise the Bellman equation exactly at the stated equalities. Every mixture over the tied actions therefore constitutes a stationary equilibrium strategy. The value function remains unique by Proposition 1. If strategies can depend on calendar time, an actor that is exactly indifferent can also use non-stationary stopping rules. Restricting attention to stationary strategies excludes these additional forms of indeterminacy and leaves the continuation value unchanged.

B.5 Derivations for Proposition 3

From equation (14),

$$\frac{X}{c} = \frac{\chi \alpha(i_h) S}{c}. \tag{59}$$

The relevant derivatives are

$$\frac{\partial (X/c)}{\partial S} = \frac{\chi \alpha(i_h)}{c} > 0, \tag{60}$$

$$\frac{\partial (X/c)}{\partial \chi} = \frac{\alpha(i_h) S}{c} > 0, \tag{61}$$

$$\frac{\partial (X/c)}{\partial i_h} = \frac{\chi S}{c} \alpha'(i_h) \geq 0, \tag{62}$$

$$\frac{\partial^2 (X/c)}{\partial S \, \partial i_h} = \frac{\chi}{c} \alpha'(i_h) \geq 0. \tag{63}$$

Using $D = 1 - \beta(1 - \pi)$, differentiation of the three cut-offs gives

$$\frac{\partial R}{\partial \pi} = 0, \qquad \frac{\partial P}{\partial \pi} = -\frac{(1-\beta)^2 c}{\beta \pi^2} < 0, \tag{64}$$

$$\frac{\partial A}{\partial \pi} = -\frac{(1-\beta) c}{\beta \pi^2} < 0, \qquad \frac{\partial \rho^*}{\partial \pi} = \frac{1-\beta}{D^2} > 0. \tag{65}$$

Differentiation with respect to relative political power gives

$$\frac{\partial R}{\partial \rho} = -\frac{(1-\beta)c}{\beta \rho^2} < 0, \qquad \frac{\partial A}{\partial \rho} = \frac{c}{\rho^2} > 0, \qquad \frac{\partial P}{\partial \rho} = 0. \tag{66}$$

Away from indifference, the equilibrium action in state h is

$$I_h^* = \mathbf{1}\{X > R(\rho)\}. \tag{67}$$

The action in state ℓ can be written as

$$I_\ell^* = \begin{cases} \mathbf{1}\{X > P\}, & I_h^* = 0, \\ \mathbf{1}\{X > A(\rho)\}, & I_h^* = 1. \end{cases} \tag{68}$$

An increase in X therefore weakly raises both actions.

Since $R(\rho)$ is independent of π, an increase in π leaves I_h^* unchanged. The relevant cut-off for I_ℓ^* is either P or $A(\rho)$, both of which decrease in π. Hence I_ℓ^* weakly increases in π.

Equation (67) and $\partial R/\partial \rho < 0$ imply that I_h^* weakly increases in ρ. For the low-state action, Proposition 2 implies

$$I_\ell^* = \begin{cases} \mathbf{1}\{X > P\}, & \rho < \rho^*, \\ \mathbf{1}\{X > A(\rho)\}, & \rho > \rho^*. \end{cases} \tag{69}$$

At $\rho = \rho^*$, $A(\rho^*) = P$. Since $A(\rho)$ is increasing in ρ, equation (69) establishes that I_ℓ^* is weakly decreasing in ρ.

If $X > P$, an equilibrium without investment is impossible. A fall in ρ can therefore preserve the equilibrium path or replace reactive investment with anticipatory investment. If $X < P$, anticipatory investment is impossible. A fall in ρ can then preserve the equilibrium path or replace reactive investment with no investment. This completes the proof.